Graduation Project Proposal

Smart OSH Performance & Risk Predictive Dashboard in Power Plants

1. Project Information

Project Title: Smart OSH Performance & Risk Predictive Dashboard in Power Plants

Course/Track: Junior Data Analyst

Team Members:

1. Alshimaa Mohamed - Team Leader
2. Mahmoud Mohamed
3. Nesma
4. Hassan

2. Project Overview

Objective

The objective of this project is to develop an interactive Power BI dashboard that monitors Occupational Safety and Health (OSH) performance indicators and helps identify potential workplace risks in power plants. By analyzing historical safety logs, incident reports, and working-condition data, the project will transform raw safety records into meaningful visual insights, classify risk levels (Low, Medium, and High) using defined safety indicators and DAX-based rules, and support safety officers in making informed, data-driven decisions.

Scope of Work

1. **Data Preparation & Modeling**
   The project covers data preparation and modeling using Microsoft Power BI. Power Query will be used to clean and transform safety data, handle missing values and duplicates, correct data types, and prepare the data for analysis. A structured data model will be designed using relationships between tables and a Star Schema where appropriate.
2. **Exploratory Data Analysis**
   The project will examine historical safety incidents, hazard categories, working conditions, and risk patterns. Power BI visuals will be used to explore trends and compare OSH performance across available categories and time periods.
3. **KPI and Risk-Level Calculation**
   DAX calculated columns and measures will be created to calculate key performance indicators (KPIs), summarize incident trends, and assign Low, Medium, or High risk categories according to agreed safety criteria. These classifications are indicator-based rules, not a trained machine-learning model.
4. **Dashboard Development & Insights Delivery**
   Interactive Power BI dashboards will present key safety metrics, historical trends, and risk classifications using visuals, filters, and drill-through capabilities where appropriate. The project will conclude with practical, data-informed recommendations to support workplace safety monitoring and decision-making.

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5. **Risk Assessment & Mitigation Plan**
   This involves identifying potential project risks and establishing practical solutions to eliminate or lower their impact on project development.

6. **Key Performance Indicators (KPIs)**
   Establishing clear metrics for project success, which include monitoring response time, ensuring smooth system updates, and measuring the user adoption rate.

Expected Outcomes

- A cleaned and transformed dataset ready for analysis in Power BI.
- A structured data model that supports clear and consistent reporting.
- A defined set of OSH KPIs calculated using DAX measures and calculated columns.
- An interactive Power BI dashboard showing historical safety trends and Low, Medium, and High risk classifications based on defined indicators.
- Actionable insights and recommendations to support safety monitoring and data-driven decisions.
- A professional project report and presentation for the graduation evaluation.

3. Problem Statement

Power plants generate safety logs, incident reports, and information about working conditions. Without structured analysis and clear visualization, these records can be difficult to use to identify recurring hazards, monitor OSH performance, and recognize areas requiring attention. The lack of a centralized dashboard may limit timely understanding of safety trends and risk indicators. Therefore, this project proposes an interactive Microsoft Power BI solution that transforms historical safety data into clear KPIs, risk classifications, and actionable insights to help safety officers make informed decisions.

4. Proposed Solution

Technologies Used

- **Microsoft Power BI:** Data visualization, interactive dashboard development, and reporting.
- **Power Query:** Data cleaning, transformation, and preparation.
- **DAX (Data Analysis Expressions):** Calculated columns, measures, KPIs, and indicator-based risk classification.
- **Microsoft Excel / CSV:** Source data handling and initial data exploration.
- **Data Modeling - Star Schema:** Organizing tables and relationships to support efficient analysis.

Proposed Dashboard Components

- **OSH Executive Dashboard:** High-level safety KPIs for management and decision-makers.
- **Incident & Hazard Analysis Dashboard:** Analysis of incident types, hazard patterns, and historical trends.
- **Risk Classification Dashboard:** Overview of Low, Medium, and High risk categories based on defined safety indicators.

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- **Safety Performance Dashboard:** Comparative views to help safety officers identify areas requiring improvement.

Team Members & Responsibilities

- **Alshimaa Mohamed - Team Leader:** Manage the project timeline, define OSH safety KPIs, interpret findings, and prepare the final DEPI documentation and presentation.
- **Mahmoud Mohamed:** Gather the datasets, review data quality, clean missing values and duplicates, and prepare the data for Power BI.
- **Nesma:** Perform exploratory data analysis and help define the measures and criteria used to analyze safety trends and classify risk levels.
- **Hassan:** Design and build the final interactive Power BI dashboard.

5. System Architecture

1. OSH Executive Dashboard: Management / Decision Makers.
2. Incident & Hazard Analysis Dashboard: Safety Officers / OSH Team.
3. Risk Classification Dashboard: Safety Officers / Operational Supervisors.
4. Safety Performance Dashboard: Plant Operations / Safety Management.

6. Resources Needed

Hardware

- Laptop or computer.

Software

- **Microsoft Power BI Desktop** - Primary tool for data modeling, analysis, and dashboard development.
- **Microsoft Excel / CSV** - For initial data exploration and source data handling.
- **Power Query (built into Power BI)** - For data cleaning and transformation.
- **DAX (Data Analysis Expressions)** - For calculated columns, measures, KPIs, and indicator-based risk classifications.

7. Proposed Project Timeline

Week 1 - Project Scope and Data Preparation

- Define project scope, safety hazards, and OSH KPIs (Alshimaa).
- Find relevant power-plant safety datasets and perform initial cleaning (Mahmoud).
- Explore safety data and identify initial patterns (Nesma).
- Prepare the initial dashboard layout (Hassan).

Deliverables: Project abstract, list of OSH KPIs, cleaned CSV dataset, and initial analysis notes.

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Week 2 - Data Modeling and Initial Dashboard

- Transform data using Power Query and establish the data model (Mahmoud).
- Review safety trends and help define risk-classification criteria (Nesma).
- Create the first dashboard visuals in Power BI (Hassan).
- Draft the project documentation (Alshimaa).

Deliverables: Prepared dataset, initial data model, and first dashboard draft.

Week 3 - KPI Development and Dashboard Integration

- Create and validate DAX measures, calculated columns, and risk classifications (Nesma with Mahmoud).
- Integrate KPIs and risk classifications into the dashboard (Hassan).
- Interpret results and draft actionable safety recommendations (Alshimaa).

Deliverables: KPI calculations and interactive dashboard showing safety trends and indicator-based risk levels.

Week 4 - Testing and Final Presentation

- Polish dashboard visuals, filters, and navigation (Hassan).
- Verify calculations, data consistency, and risk-classification rules (All team members).
- Finalize the project report and presentation slides (Alshimaa).

Deliverables: Completed Power BI dashboard file, final project report, and graduation presentation slides.

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